

Arun Vats

+91 7037487200 | vatsarun58@gmail.com | LinkedIn | GitHub | HackerRank

Education

B.Tech in Computer Science and Engineering

2023 – 2027

Bharat Institute of Technology, Meerut (AKTU)

CGPA: 8.34/10

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering

Technical Skills

- **Programming Languages:** Java, JavaScript, Python, C
- **Frontend:** React.js, HTML5, CSS3, Tailwind CSS, Redux Toolkit, Vite
- **Backend:** Node.js, Express.js, REST APIs, MongoDB, Mongoose, JWT, OAuth
- **Full Stack:** MERN Stack, RESTful API Development, Authentication, Authorization
- **AI / GenAI:** Generative AI, Agentic AI, Retrieval-Augmented Generation (RAG), LLM Applications
- **Cloud and DevOps:** AWS, Docker, Kubernetes, CI/CD Pipelines, GitHub Actions
- **Databases and Caching:** MongoDB, MongoDB Atlas, Redis
- **Developer Tools:** Git, GitHub, npm, Postman, Linux

Technical Projects

Connect Project – Open Source CLI

GitHub

- Built an open-source Node.js CLI that scans, diagnoses, repairs, connects, and deployment-checks JavaScript and TypeScript full-stack projects.
- Implemented automated framework and technology detection using multi-signal confidence scoring across frontend, backend, package manager, state management, and API client configurations.
- Developed a rule-based validation and atomic repair engine with timestamped backups, dry-run support, and rollback functionality for safe configuration changes.
- Automated frontend-backend integration by generating or repairing API clients, environment variables, CORS configuration, and backend health-check endpoints.
- Implemented deployment validation for Vercel, Netlify, Railway, Render, Fly.io, Docker, and Docker Compose configurations.
- Built a runtime error diagnostic system capable of classifying logs by category and severity and providing root-cause analysis with targeted fixes.
- Organized the codebase into modular scanners, detectors, validators, repair, connection, deployment, doctor, configuration, caching, and logging components.

create-arun-backend – Open Source npm CLI

npm GitHub

- Developed and published an npm CLI that generates production-style Express.js backend applications with a scalable layered architecture.
- Implemented JWT authentication using access and refresh tokens, Google OAuth, password hashing, and cookie-based authentication.
- Designed a modular backend architecture using Controller, Service, Repository, Model, Middleware, Validation, and Utility layers.
- Integrated security middleware including Helmet, CORS, HPP, rate limiting, compression, centralized error handling, and asynchronous error handling.
- Implemented request validation with express-validator, environment validation with Zod, and structured application logging using Pino.
- Automated generation of authentication modules, database integration, middleware, environment templates, configuration, and scalable project structures through a single CLI command.
- Gained **60K+ reactions on Instagram** after showcasing the open-source project, demonstrating strong developer-community engagement.

DeepTrust Live – Full-Stack AI Platform

Live Demo GitHub

- Engineered a full-stack misinformation and manipulated-media detection platform using React.js, Node.js, Express.js, and MongoDB.
- Integrated Sightengine API for deepfake image analysis and GNews API for automated news credibility verification workflows.
- Designed modular frontend and backend architecture with separation of concerns across presentation, routing, business logic, and data-access layers.

- Designed MongoDB data models and deployed cloud-hosted database infrastructure using MongoDB Atlas.
- Independently handled system architecture, API development, database design, third-party API integration, deployment, testing, and debugging.

Open Source and Engineering

- Built and published developer-focused open-source tools through npm, focusing on backend scaffolding, project automation, configuration validation, and developer experience.
- Applied software engineering principles including modular architecture, layered design, separation of concerns, validation, centralized error handling, structured logging, and automated repair workflows.
- Developed hands-on experience with full-stack development and cloud-native technologies including AWS, Docker, Kubernetes, Redis, and CI/CD pipelines.
- Explored Generative AI, Agentic AI, and Retrieval-Augmented Generation (RAG) for building intelligent applications and automated workflows.

Achievements

- Secured **Top 10 Position** at the **Noida Hackathon** organized by Galgotias University.
- Secured **3rd Rank** in a college-wide Artificial Intelligence Competition.
- Secured **3rd Rank** in a college-wide Coding Competition.
- Ranked among the **Top 15 Teams** in Hackheist Hackathon.
- Ranked among the **Top 15 Teams** in Trikon 2.0 Hackathon.
- Participated in **5+ hackathons**, developing and presenting software solutions under strict time constraints.

Certifications

- **HackerRank Java Certification**
- **HackerRank Python Certification**